Robotics & Human-Robot Collaboration

A simple guide to differences, importance, and evolution.

Question 1

Differentiate between a machine and a robot.

Machine vs. Robot

- **Definition:** Machine is a tool to accomplish a task; Robot perceives and reacts to surroundings.
- **Components:** Robots add sensors, actuators, and controllers to standard machine parts.
- **Requirement:** Machines need human control; Robots can function independently.
- **Applications:** Both are used in industry, transportation, and healthcare.

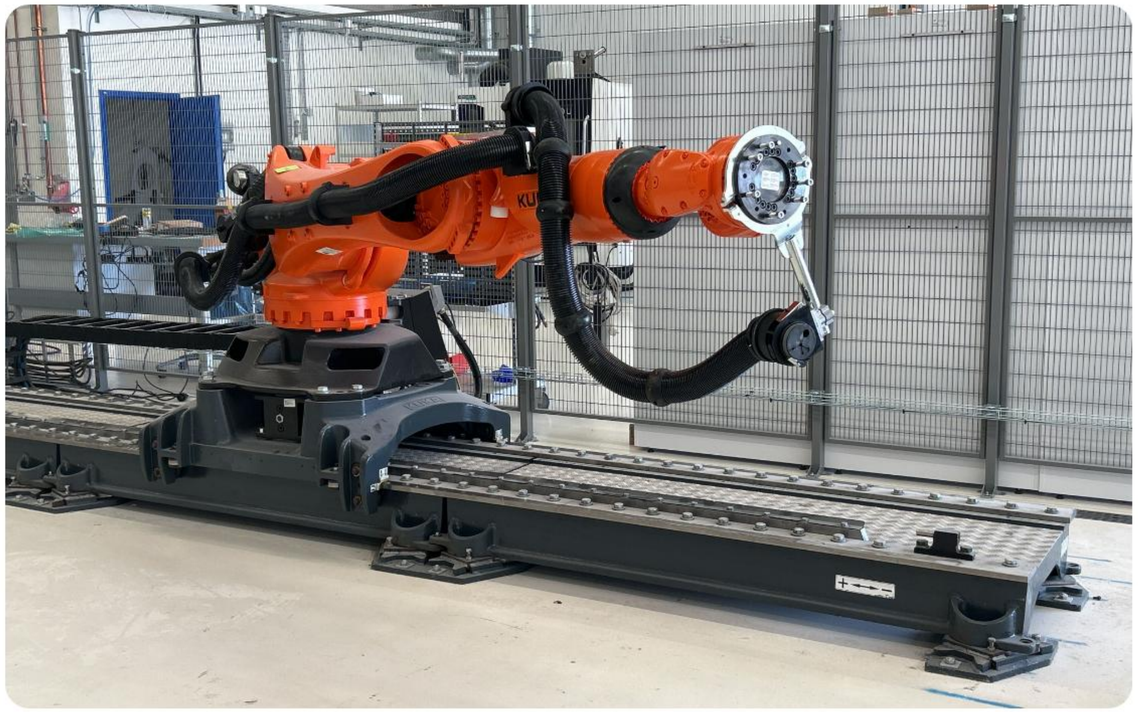


Question 2

Differentiate between a robot and a cobot.

Robot vs. Cobot

- **Purpose:** Robots automate dangerous tasks; Cobots work collaboratively with humans.
- **Design:** Robots are large and caged; Cobots are smaller, lighter, and open.
- **Performance:** Robots are fast and strong; Cobots are slower and less powerful.
- **Cost:** Robots are more expensive; Cobots are cost-effective.

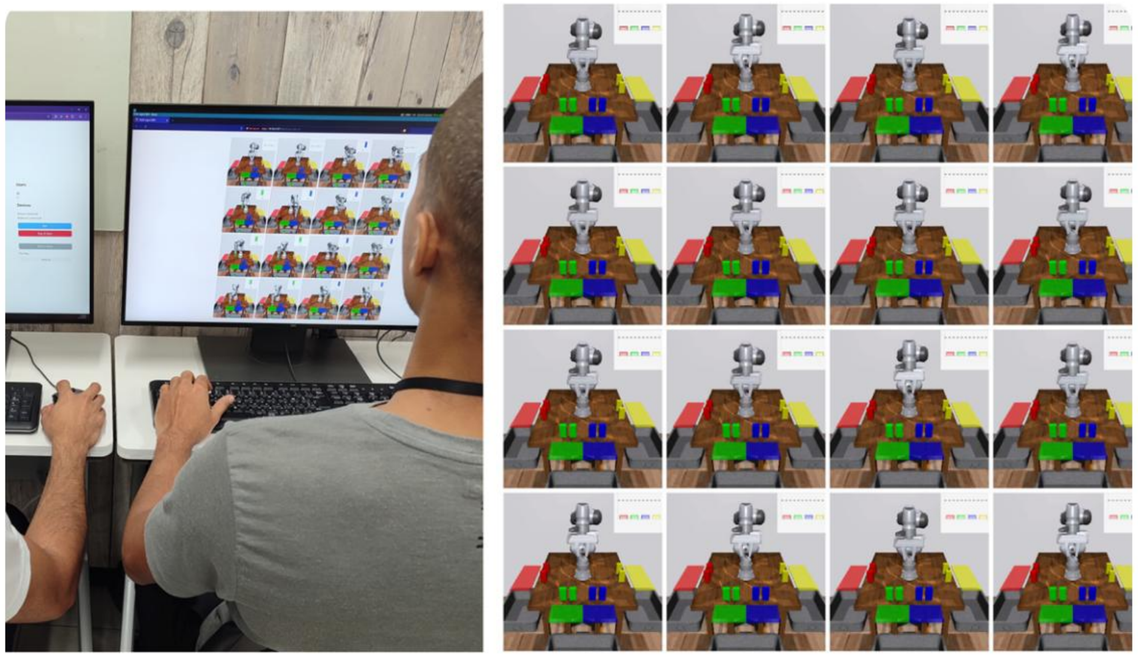


Question 3

Why is human interaction important in multi-robot systems?

Importance of Human Interaction

- • **Ethical Training:** Humans train robots for better ethical goals and behavioral flexibility.
- • **Resource Allocation:** Humans monitor and optimize deployment efficiency.
- • **Continuous Improvement:** Human feedback helps robots learn and improve over time.



s controlling 16 robots. **(b)** Close-up of the user int

Question 4

What is the motivation behind the progress from robots to cobots?

Motivation for Cobots

- **Technology:** Advancement in sensors and actuators.
- **Demand:** Growing industry demand for flexible automation.
- **Human Proximity:** Need for robots to work safely close to humans.
- **Cost/Efficiency:** Requirement for lower-cost, lighter systems.

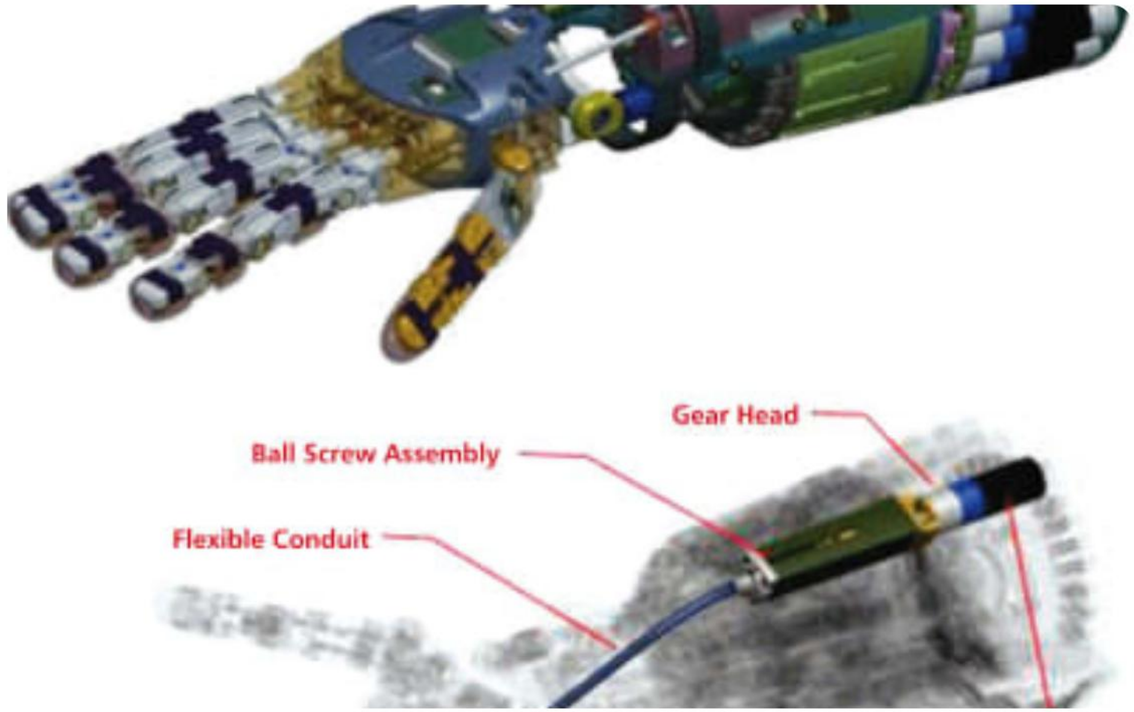


Image Sources



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